MIL-S-58004C(ME)
20 August 1984

SUPERSEDING
MIL-S-58004B(ME)
1 June 1976

MILITARY SPECIFICATION

SLING ASSEMBLY, VEHICLE LIFTING

This specification is approved for use by the Belvoir Research and Development Center, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a bridle-type sling.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir, Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3940

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SPECIFICATIONS

FEDERAL

T-R-605	- Rope, Manila and Sisal.
T-R-650	- Rope, Yarn and Twine, Bast Fiber.
FF-B-584	- Bolts, Finned Neck; Key Head; Machine; Ribbed Neck; Square Neck; Tee Head.
FF-T-276	- Thimbles, Rope.
MM-L-736	- Lumber, Hardwood.
QQ-S-781	- Strapping, Steel, and Seals.
RR-C-271	- Chains and Attachments, Welded and Weldless.
RR-W-410	- Wire Rope and Strand.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.
PPP-T-60	- Tape, Packaging, Waterproof.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-20241	- Gasket Material. Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-F-52553	- Fittings, Wire Rope.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The sling assembly, vehicle lifting (herein after referred to as "sling") shall be as shown in figure 1 and as specified herein. No deviation from the prescribed dimensions or tolerances shall be made without prior approval by the contracting officer.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The sling shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrision inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

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3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the sling may be newly fabricated from recovered materials to the maximum extent practicable, provided the sling produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the sling.

3.3.3 Wire rope. Wire rope shall be preformed right regular lay, with a fibercore, and shall conform to RR-W-410, type I, class 2, construction 3 or 4.

3.3.4 Manila rope. Rope shall be three-strand standard lay, conforming to T-R-605, type M, class 2, and shall be treated for mildew resistance.

3.3.5 Roundline. Roundline shall conform to T-R-650, type IV, class 4.

3.3.6 Lumber. Lumber for spreader bars shall conform to MM-L-736, type II, No. 1 common white oak, surface four sides (S4S), and shall be free from knots and splits.

3.3.7 Carriage bolt. Carriage bolts shall conform to FF-B-584, type I, class 1, style A. Bolts shall be provided with mated nuts and washers.

3.3.8 Anchor shackle. Anchor shackles shall conform to RR-C-271.

3.3.9 Lifting ring. Lifting rings shall conform to RR-C-271, type VI, minimum of 6 inch diameter.

3.3.10 Thimble. Thimble shall conform to FF-T-276, type III.

3.4 Construction.

3.4.1 Sling legs. A handspliced or mechanically spliced eye shall be formed over a thimble in each end of each leg. A mechanical splice shall be fabricated by pressing or swaging one or more metal sleeves over the rope junction. Swage sleeves shall conform to MIL-F-52553 or other swage sleeves that conform to the quality assurance provisions of MIL-F-52553.

3.4.2 Endless slings. The rope forming the sling shall be passed through the two bights of the jackstay, and the two ends of the rope shall be spliced together as specified in 3.4.3.

3.4.3 Nets. Two nets shall be attached to the endless slings. Nets shall be fabricated of manila rope. The outside edge of each net shall be formed by continuous jackstay around the net. A 6-inch inside perimeter bight shall be formed in each corner. The bight shall be made by seizing the two parts together at each corner of the net, a double seizing of roundline, 15 turns on the inner seizing and 14 turns on the outer seizing. The ends of the jackstay shall be joined by a five-tuck short splice consisting of three full strand

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tucks and two half-strand tucks. Before the last two strands of each tuck are made, they shall be reduced in thickness by one-half, to taper the splice. Ends of each strand shall protrude at least 3/4 inch beyond the last tuck. Meshes of nets shall be square, not more than 6 inches center-to-center. One strand of the horizontal mesh rope shall be passed under one strand of the vertical mesh rope, and evenly relaid to form a firm intersection. This shall be repeated at each intersection. The ends of mesh ropes shall be fastened to the jackstay by passing one strand of the mesh rope through the jackstay, re-laying the mesh rope, and taking a round turn about the jackstay. The end of the mesh rope shall then be spliced into its own part with three full tucks and trimmed to leave an excess of yarn equivalent to approximately two diameters of rope used.

3.4.4 Wire rope splices. Splicing of wire rope eyes shall be the full strand five-tuck splice with the core buried in the splice. Splicing shall be right-handed for regular lay wire rope with five full strand tucks. The core shall be rolled into the center of the splice. The first tuck shall be a forming tuck made by passing the dead end strand between two adjacent strands in the body of the rope and locking the dead end. Each dead end strand shall then be given four additional full tucks. Each tuck shall be made by rotating the dead end strand a full turn around a live strand in the burning. The splice shall be tapped to remove high spots and allow the strands to mesh. Each wire rope splice shall be seized with 3/32-inch seizing strand conforming to RR-W-410, type V, class 1. Seizing shall be galvanized and shall cover the splice and adjacent strands which were unlaid during splicing. Seizing shall be tightly wound, each wrap laying against the preceding wrap. The ends shall be twisted together and laid against the rope.

3.5 Lifting capacity. The sling shall lift a load of 10,000 pounds, equally distributed on the two nets without damage, deformation, or rupture of any part.

3.6 Treatment and painting. The spreader bars shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3.7 Identification marking. Identification marking of the sling shall be in accordance with MIL-STD-130. A metal tag shall be securely attached to each sling with wire. The metal tag shall be corrosion resistant and have a minimum thickness of 0.016 inch. The wire shall be corrosion-resistant and shall have a minimum diameter of 0.047 inch. The tag shall have the following information etched or embossed thereon in block letters:

1. Nomenclature: Sling Assembly, Vehicle Lifting.
2. Specification: MIL-S-58004
3. NSN: _____
4. Safe working load: _____ pounds.
5. Contract no. and date: _____.

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3.8 Workmanship. The finished slings shall be clean and free from dirt or other extraneous materials. All rope (manila and wire) shall be sound and free from severed, torn, twisted or kinked yarns or strands. Splices shall be neatly made, snuggly and securely tucked, and free of any loose strands. Splicing shall cause no severed, torn, twisted or kinked yarns or strands. Rings and shackles shall be free from cracks.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article sling shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The first article sling shall be tested as specified in 4.5.2. Failure of the test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105, inspection level S-3.

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4.4.2 Examination.

4.4.2.1 Individual. Each sling shall be examined for the critical defect specified in 4.5.1. Presence of a critical defect shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.5 percent defective for major defects and 6.5 percent for minor defects.

4.4.3 Test. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 1.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The sling shall be examined as specified herein for the following defects:

Critical

1. Cracks in rings or shackles.

Major

101. Material not as specified.
102. Materials are not resistant to corrosion and deterioration or treated to be made resistant to corrosion and deterioration for the applicable storage and operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatments.
105. Used, rebuilt or remanufactured components, pieces or parts incorporated in the sling.
106. Loose splices.
107. One or more severed yarns in fiber rope.
108. Dimensional requirements not met.
109. Splits or knots in spreader bar.
110. Components not as specified.
111. Deterioration of fiber rope.
112. Rusty, kinked, or frayed wire rope.
113. Insufficient tucks in splices.
114. Components missing.
115. Identification tag missing; markings incorrect or illegible.
116. Workmanship not as specified.

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Minor

201. Failure to taper splice.
202. Bare or thin spots on painted surface.
203. Galvanizing not complete.

4.5.2 Test. Each net of an assembled sling shall be loaded to not less than 5,000 pounds in a manner that will effectively simulate vehicle wheels. The combined weight of 10,000 pounds shall then be lifted at least 1 foot from the surface and held in suspension for 10 minutes. Nonconformance to 3.5 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of packs.

4.6.1.1 Unit of product. For the purpose of this inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

117. Nets not folded or rolled and barrier material folds, laps, or seams not sealed as specified for level A.
118. Wire rope slings not preserved as specified for level A.
119. Strapping not as specified for level A or B.
120. Bags containing the shackles and bolts not closed and secured as specified for level A.
121. Preservation and packing not in accordance with the referenced document as specified for commercial.
122. Packing not as specified for level A or B.
123. Marking missing, illegible, incorrect, or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Nets. Each net shall be folded or rolled, with the rope slings inside, to form a minimum size pack. The pack shall be secured with cotton tape or twine. The pack shall then be wrapped with barrier material conforming to PPP-B-1055, class C1 and each barrier fold, lap, and seam sealed with tape conforming to PPP-T-60, type IV.

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5.1.1.2 Wire rope sling. The sling legs shall be coated with P-1 preservative in accordance with MIL-P-116.

5.1.1.3 Spreader bars. The spreader bars shall be placed together and secured in two places with strapping. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable, and unless otherwise specified, shall be finish B (see 6.2). When specified, finish A strapping shall be used.

5.1.1.4 Shackles and carriage bolts with washers and nuts. The shackles, with pins in place, and the carriage bolts with washers and nuts, shall be preserved with P-1 and placed in a burlap or cloth bag. The bag opening shall be closed and secured by tying with soft wire.

5.1.2 Commercial. The sling shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2).

5.2.1 Level A. The components of each complete sling preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional. The sling legs shall be doubled back without kinking to form a length not greater than 10 feet. The contents shall be immobilized within the box in a manner to prevent movement. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable, and unless otherwise specified, shall be finish B (see 6.2). When specified, finish A strapping shall be used.

5.2.2 Level B. The components of each complete sling preserved as specified in 5.1, shall be packed as specified for level A except that the box shall be domestic type and strapping shall be finish A.

5.2.3 Commercial. The sling shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with ASTM D 3951. In addition, weight and cube data shall be marked on the shipping container.

6. NOTES

6.1 Intended use. The sling is intended for use in dockside loading and unloading of wheeled vehicles not exceeding 7,000 pounds.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. When a first article is required for inspection and approval and number of units required (see 3.2).

c. Degree of preservation and degree of packing required (see 5.1 and 5.2).

d. When other than finish B strapping is required (see 5.1.1.3 and 5.2.1).

6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangement for examinations, test and approval of the first article.

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Review:
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Preparing activity:
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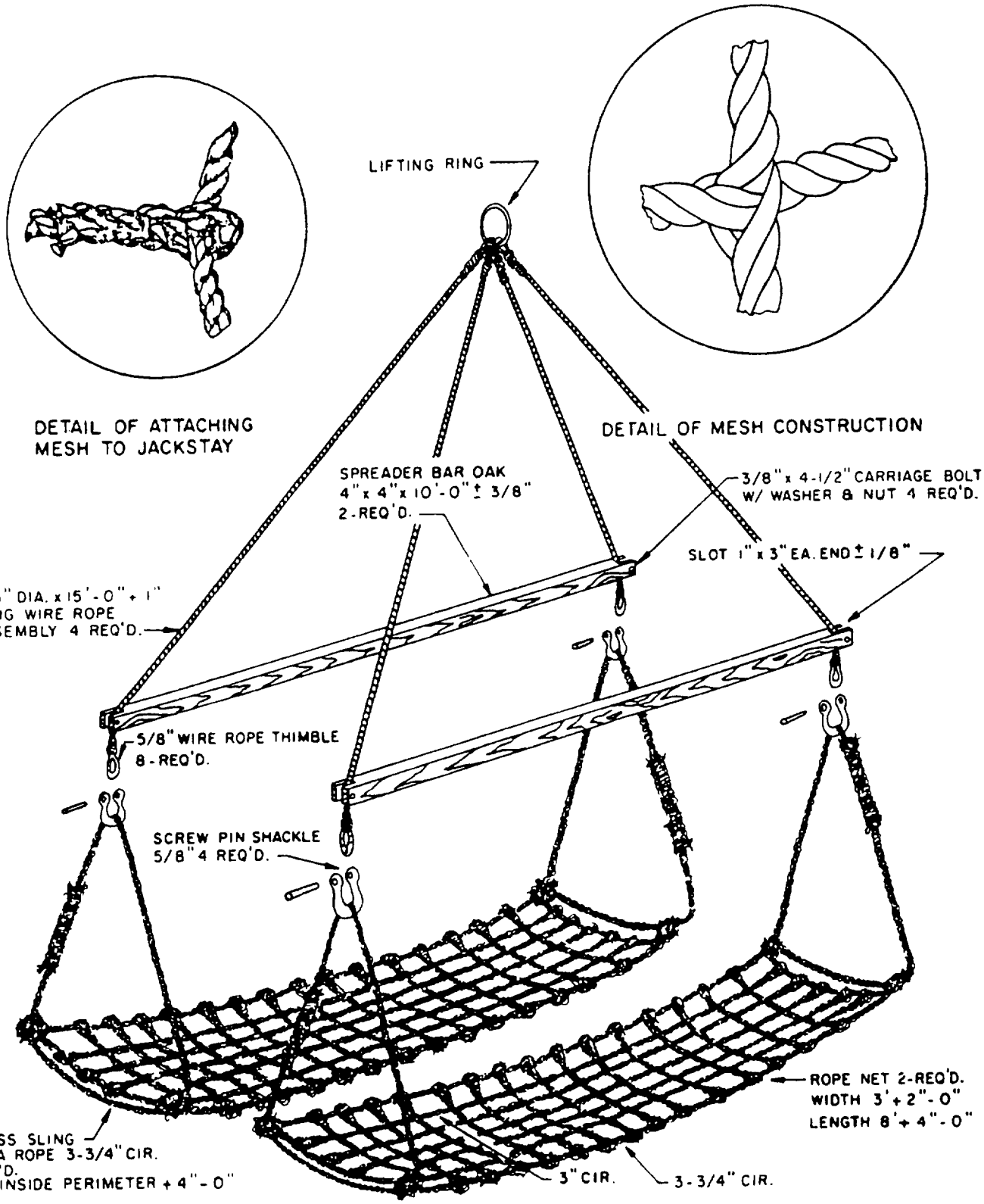


FIGURE 1. Vehicle lifting sling assembly.

X-310B

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-58004C(ME)	Sling Assembly, Vehicle Lifting

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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